Technical Assessment

3D Roof Reconstruction & Solar Panel Placement

Role: Geospatial Software Engineer — Computational Geometry & Python

1. Objective

Using the provided data, develop a solution, preferably in Python, to reconstruct rooftop geometry and generate a viable solar panel layout.

This assessment is intended to evaluate your problem-solving ability, computational geometry skills, algorithm design, and practical implementation.

2. Input Data

You will be provided with:

- DSM (Digital Surface Model)
- RGB Aerial Imagery

3. What You Need to Do

Develop a workflow that addresses the following:

- Calculate elevation, slope, and aspect/azimuth.
- Identify roof faces and reconstruct a geometrically consistent roof surface.
- Regularize noisy elevation values and handle missing data through appropriate interpolation or reconstruction.
- Identify rooftop obstructions, such as chimneys, skylights, and roof windows, and exclude them from usable areas.
- Generate a 3D representation of the reconstructed roof.
- Identify suitable roof areas and place solar panels while respecting geometric constraints, including available usable area, sufficient space for standard-sized panels, compact panel placement, and avoidance of rooftop obstructions such as chimneys, windows, and skylights.

You are free to choose your own algorithms and implementation approach.

4. Deliverables

Submit the following:

1. **Video demonstration:** A maximum of 5 minutes, demonstrating your solution and its results.
2. **Python Source Code:** Host your solution in a **public GitHub repository**. Include a well-structured README with setup instructions, dependencies, and clear steps to run the solution.
3. **Visual results:** Show the reconstructed roof, identified obstructions, and final panel placement.
4. **Brief technical explanation:** Describe your approach, key algorithms, assumptions, limitations, and any incomplete tasks.

5. Evaluation Criteria

We will assess your submission based on:

- Computational geometry and mathematical reasoning.
- Quality and consistency of the reconstructed roof.
- Handling of noisy data, missing values, and obstructions.
- Accuracy and geometric validity of solar panel placement.
- Python code quality and algorithm design.
- Validation, problem-solving, and technical decision-making.
- Clarity in presenting and explaining your approach.
- English fluency during the solution demonstration.

Important: We value the quality of your reasoning and implementation over visual appearance alone. You should be prepared to explain and defend your technical decisions, demonstrate your implementation, and answer detailed questions about your approach during the technical discussion.

Submit only work that you can explain and demonstrate yourself.

We look forward to seeing what you can create from the provided data.

Best of Luck!